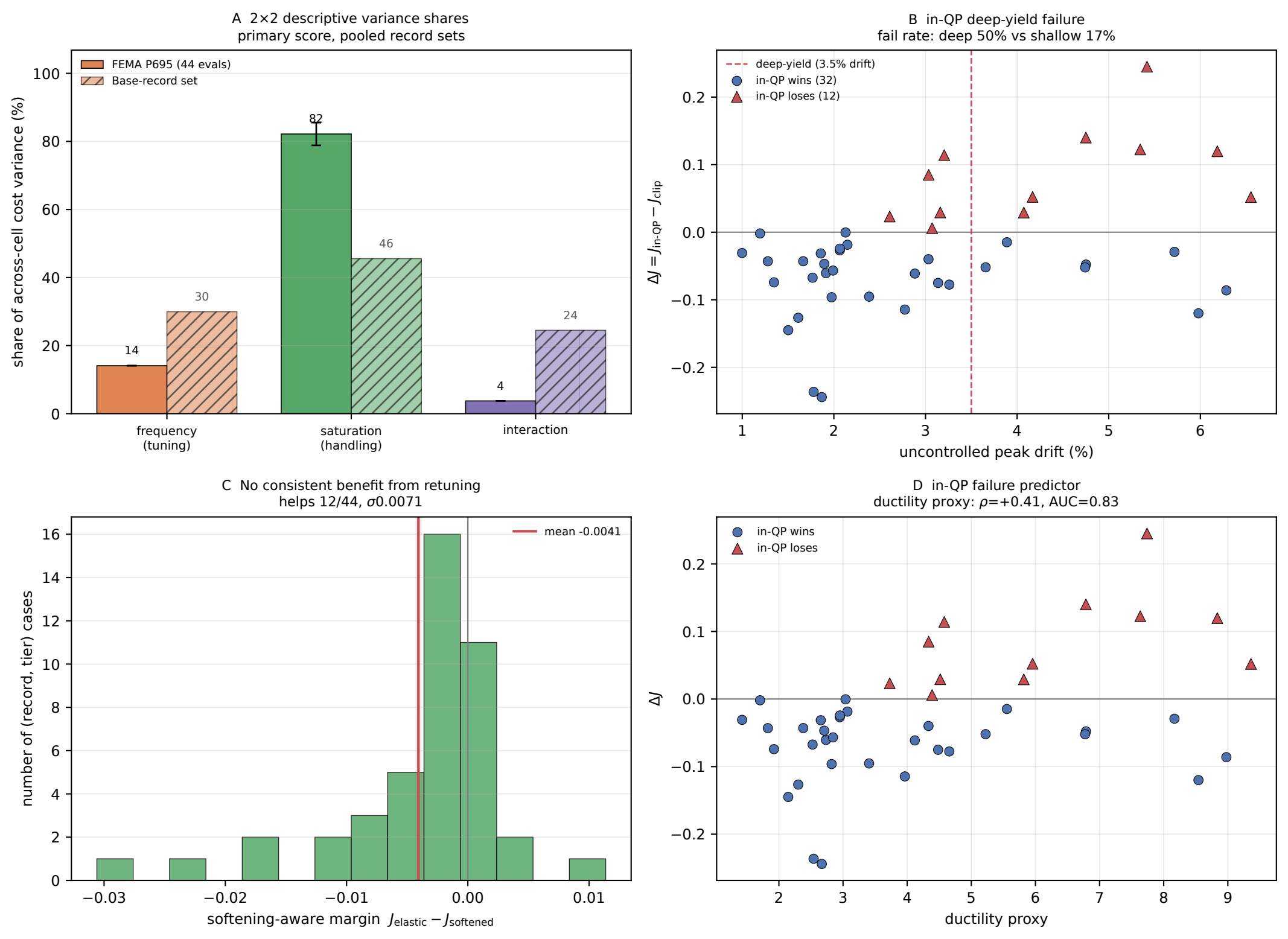

A 2×2 descriptive variance shares
primary score, pooled record sets
FEMA P695 (44 evals)
Base-record set
share of across-cell cost variance (%)
14
30
82
46
4
24
frequency (tuning)
saturation (handling)
interaction
B in-QP deep-yield failure
fail rate: deep 50% vs shallow 17%
deep-yield (3.5% drift)
in-QP wins (32)
in-QP loses (12)
$\Delta J = J_{\text{in-QP}} - J_{\text{clip}}$
uncontrolled peak drift (%)
C No consistent benefit from retuning
helps 12/44, σ0.0071
mean -0.0041
number of (record, tier) cases
softening-aware margin $J_{\text{elastic}} - J_{\text{softened}}$
D in-QP failure predictor
ductility proxy: ρ=+0.41, AUC=0.83
in-QP wins
in-QP loses
ΔJ
ductility proxy